

Appendix: Methodology for the Conjoint Analysis Question

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1 Overview

This appendix describes in full detail the construction of the conjoint analysis question included in the Global Justice questionnaire. The conjoint presents respondents with pairs of stylised societal scenarios and asks them to choose between them. Each scenario is characterised by eight underlying parameters (Section 3) and displayed through seven features (Sections 7–9). The income distributions underlying the features are derived from Bothe et al. (2026) and Chancel et al. (2026); temperature projections rely on the emission pathways from Chancel et al. (2026). All monetary quantities are expressed in EUR PPP 2025 per adult per year unless otherwise noted.

2 Scenario nomenclature

A scenario is characterised by four structural dimensions:

Hours (or class). The average weekly working hours converge to a 2100 target corresponding to one of: S15k, S45k, S (= S60k), B (= S at 2100, constant-growth productivity), B90k (formerly M), B120k (formerly W), P (from shortest to longest; see Table 1). B has the same 35h/60k target as S but uses constant-growth productivity at 2035 (see Section 5.6).

Scope. The redistribution configuration: C = Convergence (both national and global); G = Global only; N = National only; I = Inequality (neither).

Sectoral change. 1 = no sectoral change (stable public services, current food system); 2 = structural change (expanded public services, reduced beef and flights).

Decarbonization. SD = slow; ID = intermediate; FD = fast.

By convention, the scope is embedded in the hours label (e.g. SC45k2_FD for 30 h/wk, Convergence, structural change, fast decarbonization). The default benchmark is SC2_FD (35 h, Convergence, structural change, fast decarbonization).

Table 1: Mapping between working hours and 2100 GDP per capita (k€/adult).

Weekly hours	Hours label	2100 GDP pc (k€)	
		With global redistribution	Without global redistribution
29	B30k	30	25
30	B45k	45	37.5
35	S / B	60	50
40	B90k	90	75
45	B120k / P	120	100

3 Eight underlying parameters

Each respondent profile is defined by one parameter representing the respondent themselves and seven policy-relevant parameters:

1. **Individual income:** the respondent’s household monthly cash income and whether they live as a couple. The annual individual income is $y_0 = \text{HH_monthly} \times 12/k$, where $k = 2$ if couple, $k = 1$ otherwise.
2. **Decarbonization:** Slow (SD), Intermediate (ID), Fast (FD), corresponding to IEA CPS, STEPS and NZE pathways respectively. The income cost at 2035 is 0 % (SD), 1 % (ID) or 3 % (FD).
3. **Working hours:** each scenario class converges to a 2100 GDP-per-capita target (labels 29/30/35/40/45, i.e. 30/45/60/90/120 k€; see Table 1). The corresponding weekly hours *worked in 2035* are 28/32/36/40/44, evenly spaced around the B90k baseline (40 h) in steps of 4.
4. **National redistribution:** current level, or National redistribution (N).
5. **Global redistribution:** current level, or Global redistribution (N).
6. **Public services:** stable (20 % of GNI) or increased (+3 percentage points of GNI, financed by a flat-rate tax).
7. **Beef and flights:** no change; −60 % beef; −50 % air travel; both changes.

The full design yields $5 \times 2 \times 2 \times 3 \times 2 \times 4 = 480$ unique scenarios.

4 Income distribution data sources

4.1 National secondary income from [Bothe et al. \(2026\)](#)

The [Bothe et al. \(2026\)](#) simulation provides, for each country c , gpercentile p and year t :

- $\text{sdiinc}_{c,p,t}$: the income share of bracket p in country c ;
- $\text{nni}_{c,t}$: national net income per adult (EUR PPP 2025);
- $\text{diff}_{c,t}$: a normalising scalar;
- $\text{ypt}_{c,p,t}$: the global income tax (GIT) paid;
- div_t : the uniform global dividend received by every adult.

The resulting per-adult full income is:

$$\text{income}_{c,p,t} = \underbrace{\frac{\text{sdiinc}_{c,p,t} \cdot \text{nni}_{c,t}}{\text{diff}_{c,t}}}_{\text{yp_recomp: national secondary income, pre-GIT}} - \text{ypt}_{c,p,t} + \text{div}_t. \quad (1)$$

yp_recomp is the national secondary income *after* all national taxes and transfers but *before* the global income tax. All four redistribution scopes are therefore defined as:

$$\text{SC}_{c,p,t} = \text{income}_{c,p,t} = \text{yp_recomp} - \text{ypt} + \text{div}, \quad (2)$$

$$\text{SN}_{c,p,t} = (\text{SC}_{c,p,t} - \text{GR}_t) \left(G_c^{SI_0} / G_c^{SC} \right), \quad (3)$$

$$\text{SI}_{c,p,t} = \frac{1}{2} \text{PI}_{c,p,t} \quad (\text{in 2100; see Section 6}), \quad (4)$$

$$\text{SG}_{c,p,t} = \text{SI}_{c,p,t} \left(1 - \tau^G(\text{SI}_{c,p,t}) \right). \quad (5)$$

where $G_c^{SC}, G_c^{SI_0}$ are the gp-weighted mean incomes of the SC and SI ($= \frac{1}{2}\text{PI}$) distributions and $\text{GR} = \text{div} - \text{ypt}$ is the net global transfer (so $\text{SC} - \text{GR} = \text{yp_recomp}$). The N scope rescales the nationally-redistributed income to the no-global GDP level G^{SI_0} . The G scope keeps the SI (no-redistribution) within-country shape and applies a *global* tax schedule that is a function of N (national, pre-global) income, $\tau^G(\cdot)$: at world percentile g the N income $\text{SN}_g^{\text{World}}$ faces the rate $(\text{SN}_g^{\text{World}} - \text{SC}_g^{\text{World}}) / \text{SN}_g^{\text{World}}$ (negative for poor incomes, positive for rich). This schedule is then *evaluated at each unit's own I (SI) income*. Because τ^G is read off the *world* distribution, the identity $\text{SG} = \text{SI}(\text{SC}/\text{SN})$ holds at the world level but not within a single country.

Point of attention: [Bothe et al. \(2026\)](#) provide a single SC simulation. The SC1 and SC2 variants (sectoral change off/on) and all decarbonization paces (SD/ID/FD) share the same income distribution. The differences between them are modelled *explicitly* by us through (a) an income-cost multiplier for decarbonization (Section 5.3) and (b) a flat-rate PS tax for structural change (Section 5.4).

4.2 Cash income for Italy 2025

The conjoint uses *gross cash income*: primary income at market prices, net of all taxes (including VAT) and net of imputed rents and retained earnings, but *gross* of capital

consumption. This corresponds to what a respondent would recognise as their disposable income plus employer social contributions, comparable across scenarios.

Starting from [Fisher-Gethin and Chancel \(2023\)](#) data for Italy:

$$\text{cash}_p = \left(a_p^{\text{pre}} - \tau_p^{\text{dir}} - \tau_p^{\text{soc}} - \tau_p^{\text{ind}} + g_p^{\text{soc}} \right) \text{NNI} - r_e \kappa_p \text{NNI} - r_h H_p \text{NNI} + \kappa_p \text{CFC} - \text{ypt}_p, \quad (6)$$

where:

- a_p^{pre} = normalised pretax income share at percentile p ;
- $\tau_p^{\text{dir}}, \tau_p^{\text{soc}}, \tau_p^{\text{ind}}$ = direct taxes (PIT + wealth + CIT), social contributions, and indirect taxes (VAT), each normalised by mean pretax income;
- g_p^{soc} = government cash transfers;
- $r_e = 0.50 \times \bar{m}$ = retained-earnings deduction rate ($\bar{m}$ = WID corporate profit retention ratio for Italy);
- $r_h = 0.035$ = imputed-rent yield on net housing wealth ([Garbinti et al., 2021](#));
- H_p = normalised housing-wealth allocation at percentile p (interpolated from the 2014 French wealth-composition data of [Garbinti et al. 2021](#));
- κ_p = capital-income share at percentile p ;
- CFC = capital consumption per adult (so as to gross up from NNI to GNI for the capital share);
- ypt_p = GIT tax already paid in 2025.

We enforce monotonicity on the bottom-99 % of the distribution by sorting values in ascending order, mirroring the procedure in the main distributional series. All values are expressed in EUR PPP 2025.

Point of attention: τ^{ind} (VAT) is subtracted because the [Fisher-Gethin and Chancel \(2023\)](#) pretax income is at market prices (inclusive of indirect taxes). Subtracting VAT converts to factor-cost NNI, to which cash incomes sum. Omitting this step would overstate cash income.

5 IT distributions at 2035

5.1 Base distribution: IT35

Let IT35_p denote the gross cash income distribution for Italy at 2035 under SC1_SD (slow decarbonization, no structural change). It serves as the universal base from which all 2035

IT scenario distributions are derived. We compute it by rescaling the 2035 [Bothe et al. \(2026\)](#) income distribution to cash units using the 2025 individual cash-to-income ratio:

$$\text{IT35}_p = \text{income}_{IT,p,2035} \times \frac{\text{cash}_p}{\text{income}_{IT,p,2025}}. \quad (7)$$

This per-percentile ratio preserves the distributional shape while anchoring the level to the respondent-observable 2025 cash income.

5.2 GR35: the 2035 net global redistribution transfer

The per-percentile net global redistribution transfer in 2035 is

$$\text{cash_GR35}_p = (\text{div}_{2035} - \text{ypt}_{p,2035}) \times \frac{\text{cash}_p}{\text{income}_{IT,p,2025}}. \quad (8)$$

cash_GR35_p is positive for low-income individuals (net GIT receivers) and negative for high-income individuals (net GIT contributors).

Point of attention: Monotonicity is *not* enforced on cash_GR35 . Unlike income distributions, which must be weakly increasing, the net GIT transfer is mechanically decreasing with income (higher earners pay more than they receive). Sorting it would scramble the distributional signal.

5.3 Decarbonization income factor

We model the income cost of decarbonization as a flat multiplicative factor applied to all 2035 distributions:

$$\delta = \begin{cases} 1.00 & (\text{SD}) \\ 0.99 & (\text{ID}) \\ 0.97 & (\text{FD}) \end{cases} \quad (9)$$

This captures a -3% income penalty for fast decarbonization relative to slow, consistent with the income-side impact of accelerated investment in the energy transition.

Point of attention: The factor δ is applied *on top of* the [Bothe et al. \(2026\)](#) income distribution. [Bothe et al. \(2026\)](#) report the same income distribution regardless of the decarbonization pace (the SD, ID and FD variants differ only in emissions and temperature). The δ multiplier is therefore an explicit parametric cost not already embedded in the simulation.

5.4 Public-services flat-rate tax

When public services are expanded (`food=2`), the additional expenditure is financed by a flat-rate tax at rate

$$\tau_{\text{PS}} = \frac{(s_{2035} - s_{2025}) \times \bar{g}_{2035}}{\text{cash}_{2035}}, \quad (10)$$

where s_t = share of educ+health+PS spending in GNE (from [Chancel et al. \(2026\)](#) sheet **G5s**), $\bar{g}_{2035}$ = per-adult GNE at 2035 (**G0p**), and cash_{2035} = average of IT35. For Italy: $\tau_{\text{PS}} \approx 4.9\%$.

5.5 Scenario-specific 2035 IT distributions

All 2035 IT columns are derived from IT35 using the following formulas, where $\delta_{\text{FD}} = 0.97$ and $\psi = 1 - \tau_{\text{PS}}$:

$$\begin{aligned}
\text{IT35} &= \text{IT35}_p & (\text{SC1_SD: base}) & (11) \\
\text{SCmat} &= \text{IT35}_p \cdot \delta_{\text{FD}} & (\text{SC1_FD}) & (12) \\
\text{SC} &= \text{IT35}_p \cdot \delta_{\text{FD}} \cdot \psi & (\text{SC2_FD}) & (13) \\
\text{PC} &= 1.15 \text{IT35}_p \cdot \delta_{\text{FD}} & (\text{PC1_FD}) & (14) \\
\text{PI} &= 1.4 \text{IT25}_p \cdot \delta_{\text{FD}} & (\text{PI1_FD}) & (15) \\
\text{SC45k} &= (G_{2025}/G_{2035}) \text{IT35}_p \cdot \delta_{\text{FD}} \cdot \psi & & (16) \\
\text{SC15k} &= 0.9 (G_{2025}/G_{2035}) \text{IT35}_p \cdot \delta_{\text{FD}} \cdot \psi & & (17) \\
\text{SI} &= (G_{2035}^{PI}/G_{2025}) (h_{2035}^{SC}/h_{2035}^{PI}) \text{IT25}_p \cdot \delta_{\text{FD}} \cdot \psi & & (18) \\
\text{SN} &= (\text{IT35}_p - \text{cash_GR35}_p) (G^{SI_0}/\overline{\text{SC}}) \cdot \delta_{\text{FD}} \cdot \psi & & (19) \\
\text{SG} &= (\overline{\text{SC}}/G^{SI_0}) (\text{SI0}_p + \text{cash_GR35}_p) \cdot \delta_{\text{FD}} \cdot \psi & & (20) \\
\text{B90kC} &= \Phi \text{IT35}_p \cdot \delta_{\text{FD}} \cdot \psi & (\text{B90kC2_FD: 40h baseline} + \text{GIT; formerly } & (21) \\
& & & (22) \\
\text{B120kC} &= \frac{44}{40} \Phi \text{IT35}_p \cdot \delta_{\text{FD}} \cdot \psi & (\text{B120kC2_FD: 45h class, 44h worked; at 210} & (22) \\
& & & (23) \\
\text{B90kMat} &= \Phi \text{IT35}_p \cdot \delta_{\text{FD}} & (\text{B90kC1_FD: no PS tax; formerly MCmat}) & (23) \\
& & & (24) \\
\text{B} &= \frac{36}{40} \Phi \text{IT35}_p \cdot \delta_{\text{FD}} \cdot \psi & (\text{B2_FD: 35h class, 36h worked; at 2100 = SC} & (24) \\
& & & (25) \\
\text{Bmat} &= \frac{36}{40} \Phi \text{IT35}_p \cdot \delta_{\text{FD}} & (\text{B1_FD: B without PS tax}) & (25) \\
& & & (26) \\
\text{Bbeef} &= \text{B} & (\text{B2_FD, beef focus; income = B}) & (26) \\
& & & (27) \\
\text{Bflights} &= \text{B} & (\text{B2_FD, flights focus; income = B}) & (27) \\
& & & (28) \\
\text{B45kC} &= \frac{32}{40} \Phi \text{IT35}_p \cdot \delta_{\text{FD}} \cdot \psi & (\text{B45kC2_FD: 30h class, 32h worked; formerly} & (28) \\
& & & (29) \\
\text{B30kC} &= \frac{28}{40} \Phi \text{IT35}_p \cdot \delta_{\text{FD}} \cdot \psi & (\text{29h class, 28h worked; formerly MC30k}) & (29) \\
& & & (30) \\
\text{B15kC} &= \frac{24}{40} \Phi \text{IT35}_p \cdot \delta_{\text{FD}} \cdot \psi & (\text{28h class, 24h worked; formerly MC15k}) & (30)
\end{aligned}$$

where G_t denotes Italy's per-capita GDP at year t (Chancel et al. 2026 A0p; G_{2035}^{PI} is the PI per-capita GDP, A0pi); $h_{2035}^{SC}, h_{2035}^{PI}$ are per-capita economic labour hours under SC/PI (Chancel et al. 2026 E0h/E0k); $\text{SI0}_p = (G_{2035}^{PI}/G_{2025})(h^{SC}/h^{PI}) \text{IT25}_p$ is the SI base,

with $\overline{SC}$, G^{SI_0} the gp-weighted means of the SC and SI bases; and Φ is the B90k income scale factor defined in Section 5.6. B90k (40 h) is the baseline: every other B class equals Φ IT35_p times its 2035 worked-hours ratio (28/32/36/40/44 over 40).

Point of attention: PI uses IT25 (2025 cash income) as its base, not IT35, because the PI scenario does not undergo the SC productivity and redistribution transition between 2025 and 2035. It grows from the current distribution by a flat GDP-per-capita factor of 1.4. By contrast, SC and its variants start from IT35, which already embeds the [Bothe et al. \(2026\)](#) SC income structure at 2035.

Point of attention: SI uses IT25 as its base, scaled to the no-global GDP level: the PI per-capita GDP adjusted for SC's shorter working hours, $(G_{2035}^{PI}/G_{2025})(h^{SC}/h^{PI})$. It preserves the current within-country shape (no redistribution). SN and SG then sit at the same no-global level G^{SI_0} (SN) and the converged level $\overline{SC}$ (SG): SN rescales the SC-redistributed income (IT35 – GR35) to G^{SI_0} , and SG rescales the global-transfer-augmented SI base (SI0 + cash_GR35) to $\overline{SC}$.

5.6 B90k scenario: moderate convergence (formerly MC)

The B90k scenario (formerly M/MC) holds per-*worker* working hours constant at their 2025 level (40 h) and is the baseline of the B family. Its 2035 income scale relative to IT35 is:

$$\Phi = \frac{(1 + g)^{10}}{\omega_{2035} \cdot \phi_{2035}}, \quad g = 1\%, \quad (31)$$

where $(1 + g)^{10}$ is the cumulative 2025–2035 income growth of the B scenarios at a flat yearly rate $g = 1\%$ (this replaces the former practice of using the average 2025–2100 productivity rate $\bar{g}^{10}$, with $\bar{g} = (125/\phi_{2025})^{1/75}$, which front-loads SC's high near-term growth), $\phi_{2035} = \phi_{2035}/\phi_{2025}$ is the actual SC productivity growth 2025–2035 (from [Chancel et al. \(2026\)](#) sheet F0a), and $\omega_{2035} = h_{2035}^{pw}/h_{2025}^{pw}$ is the SC per-*worker*-hours change 2025–2035 (sheet E0a).

At 2100, B90k income is a fixed multiple of SC: $B90kC_{2100} = 1.5 \times SC_{2100}$ (the 90k GDP-per-capita target, vs SC's 60k), applied uniformly to IT and the world distribution. The 2100 hours ladder uses fixed GDP-target coefs (target/60) for all hours levels (see Table 2); the per-worker hours ratio from sheet E0a is used only for the 2035 income scale (below).

Point of attention: In the SC scenario, productivity growth is front-loaded (peaking around 2080 before declining), so the actual SC growth ϕ_{2035} can exceed the flat-rate numerator $(1.01)^{10} \approx 1.105$. In that case $\Phi < 1$, meaning B90k income at 2035 is *lower* than SC income — a plausible result because SC extracts peak productivity via its 35 h week while B90k works 40 h at a lower per-hour frontier.

6 Income distributions at 2100

6.1 Income concept and rescaling

At 2100, within-country income comparisons across scenarios and between countries are best made using “full income” (national secondary income inclusive of all public services), rather than cash income. However, to keep the 2100 income comparable with the 2035 cash income entered by the respondent (and shown in the IT35/SC columns), we rescale by a *single scalar*:

$$r_{IT} = \frac{\overline{\text{cash}}_{IT,2025}}{\overline{\text{income}}_{IT,2025}}, \quad (32)$$

where both averages are taken over the 127-bracket Italy 2025 distribution weighted by bracket widths. In the data $r_{IT} \approx 0.69$.

This ratio is applied uniformly across all gpercentiles and all countries. The resulting 2100 distributions therefore represent cash-income-equivalent magnitudes, stripped of the within-country variation in the cash-to-income ratio. The consistency check is: $r_{IT} \times \bar{y}_{IT,2100} \approx r_{IT} \times \bar{y}_{World,2100}$ in the SC scenario (since SC achieves near-complete global convergence by 2100), which holds in the data.

Point of attention: The decision to use a *scalar* r_{IT} for 2100 (rather than the per-percentile ratio used for 2035) is deliberate. For 2035, the per-percentile ratio is needed to preserve distributional comparability with the cash income entered by the respondent. For 2100, cross-country and cross-scenario comparability is the priority, which requires a *uniform* rescaling so that the shape of the 2100 world distribution reflects true inequality rather than the Italy-specific cash-to-income heteroscedasticity.

6.2 IT 2100 distributions

For Italy at 2100, in raw full-income units (before r_{IT}):

$$SC_{IT,2100} = \text{income}_{IT,p,2100} \quad (\text{Bothe et al. 2026 SC: GIT+dividend}) \quad (33)$$

$$SN_{IT,2100} = (\text{income}_{IT,p,2100} - \text{div}_{2100}) (G_{IT}^{SI_0}/G_{IT}^{SC}) \quad ((SC-GR) \times G^{SI_0}/G^{SC}) \quad (34)$$

$$PI_{IT,2100} = \text{yp_recomp}_{IT,p,2025} \times g_{IT}^{PI} \quad (g^{PI} = A0pi_{2100}/A0pi_{2025}) \quad (35)$$

$$SI_{IT,2100} = \frac{1}{2} PI_{IT,2100} \quad (36)$$

$$SG_{IT,2100} = SI_{IT,2100} (1 - \tau^G(SI_{IT,2100})) \quad (\tau^G \text{ a function of N income, read at IT's SI income}) \quad (37)$$

For all hours variants, apply the coefs from Table 2 to the SC distribution:

$$X_{IT,2100} = c_X \times SC_{IT,2100}. \quad (38)$$

Each weekly-hours level maps to a fixed 2100 GDP-per-capita target (k€, with global redistribution), so its coef is simply target/60: B30k (29h, 30k)→ 0.5, B45k (30h, 45k)→ 0.75, B (35h, 60k)→ 1.0, B90k (40h, 90k)→ 1.5, B120k (45h, 120k)→ 2.0. B (formerly M60k) and Bbeef/Bflights/Bmat all equal SC at 2100 (B reaches the SC level at 2100).

Table 2: 2100 income coefs relative to SC. Applied to both IT and World distributions. B food variants (Bbeef, Bflights, Bmat) equal SC at 2100 (B=SC at 2100; food does not affect 2100 income). B120k and P share the same 2100 coef (2.0); they differ only in their 2035 income formula.

Scenario	B15kC	B30kC	SC (=B,Bmat,Bbeef,Bflights)	B45kC	B90kC	B120kC, PC
Hours	28h	29h	35h	30h	40h	45h
GDP target (k€)	15	30	60	45	90	120
Coef	0.25	0.50	1.00	0.75	1.50	2.00

Point of attention: PI base uses yp_recomp_{2025} (pre-GIT national secondary income), not income_{2025} (which includes GIT transfers). The PI scenario has no global income tax; using income_{2025} would inflate the PI distribution for net GIT receivers and depress it for net contributors.

Point of attention: SCmat and SC have identical income distributions at 2100. Structural change affects emissions and temperature (and therefore 2035 incomes via δ) but not 2100 full incomes: by 2100, both modalities converge to the same equilibrium income level in Bothe et al. (2026)'s simulation.

6.3 World 2100 distributions

The world distribution is built by pooling all (country, gpercentile) cells, weighting each cell by $w_{c,p} = \Delta_p \times \text{Pop}_c$, sorting by income value, and binning into 100 equal-population global percentiles. Cells with non-positive or non-finite income are dropped. Gaps in the resulting 100-grid are filled by linear interpolation.

Populations used follow whether the scenario has global redistribution (GIT): Pop^{SC} (Chancel et al. 2026 sheet Z0a, ≈ 9.41 B in 2100, Convergence) for SC, SG, SC45k, SC15k, B90k and PC; and Pop^{PI} (Z0b, ≈ 10.18 B, Inequality) for PI, SI and SN.

Point of attention: PC inherits the SC world distribution scaled by 2.0, using SC population. A more rigorous treatment would use the UN-medium population for PC (as Chancel et al. 2026 recommend), but the implied distributional error is small since country-level growth differences matter more than aggregate population.

6.4 Output format for ineq_2100

The file `ineq_2100.csv` contains 21 income brackets: $p_0p_5, p_5p_{10}, \dots, p_{90}p_{95}, p_{95}p_{99}, p_{99}p_{100}$. Each IT bracket is the gp-width-weighted mean over the 127-grid; each World bracket is the unweighted mean over the corresponding 100-grid bins. All values are rounded to the nearest integer EUR.

7 Temperature at 2100

7.1 Base temperature

Temperature depends on GDP (which encodes working hours and global redistribution), decarbonization pace, and sectoral change. For scenarios present in Chancel et al. (2026)'s `Emission_Output` workbook, we use T_{final} from `chancel_temp2100_completed.csv`. For scenarios not modelled by Chancel et al. (2026) (SI, SN, SG, B90k and their variants), we predict temperature using a GDP-based no-emissions regression estimated on the 24 scenarios observed by Chancel et al. (2026):

$$T = \beta_0 + \beta_1 G + \beta_2 \mathbf{1}_{\text{ID}} + \beta_3 \mathbf{1}_{\text{SD}} + \beta_4 G \mathbf{1}_{\text{ID}} + \beta_5 G \mathbf{1}_{\text{SD}} + \beta_6 c \mathbf{1}_{\text{FD}} + \beta_7 c \mathbf{1}_{\text{ID}} + \beta_8 c \mathbf{1}_{\text{SD}} + \varepsilon, \quad (39)$$

where G = total 2100 GDP (T€), $\mathbf{1}_d$ = indicator for decarbonization pace d , $c = \mathbf{1}(\text{structural change} = 1)$, and ε is the residual. Adjusted $R^2 \approx 0.997$.

The total 2100 GDP is computed as $G = g_{\text{pc}} \times P$, where g_{pc} is GDP per capita from Table 1 and P is the 2100 population (9.41 B for SC family, 10.18 B for PI/no-GIT at 45h).

7.2 Temperature adjustments for beef, flights and public services

Following [Chancel et al. \(2026\)](#), once a “canonical” modality is fixed, additive corrections are applied:

- **Beef reduction:** $-0.24\text{ }^{\circ}\text{C}$.
- **Flight reduction:** $-0.155\text{ }^{\circ}\text{C}$.
- **Expanded public services** (relative to no expansion): $\pm 0.0004875 \times (67.25 + 2.06 g_{\text{pc}})\text{ }^{\circ}\text{C}$, where g_{pc} is in $\text{k}\text{€}/\text{adult}$. This formula is derived from the regression of fossil emissions on GDP per capita across SC scenarios (Section 17 of the `questionnaire_legacy.R` script).

The canonical modality is chosen as follows ([Chancel et al., 2026](#), `method_questionnaire.md`):

SD and ID: Fix the change according to the public-services choice (stable $\rightarrow$ 1, increased $\rightarrow$ 2). Adjust for beef and flights relative to the canonical co-variation with structural change.

FD: Fix the change according to the beef choice (no reduction $\rightarrow$ 1, beef reduced $\rightarrow$ 2). Land-use effects dominate at fast decarbonization. Adjust for flights and public services (the latter already captured via the sectoral-change dummy in equation 39).

Point of attention: Temperature is rounded to the nearest $0.1\text{ }^{\circ}\text{C}$. In the no-emissions regression the maximum absolute deviation from the temperatures observed by [Chancel et al. \(2026\)](#) is $0.1\text{ }^{\circ}\text{C}$ (by construction: the model was selected to guarantee this on the training set). For extrapolated scenarios (SI/SN/SG/MC) the true error is unknown but bounded by the regression’s generalization uncertainty.

8 Own income at 2035

1. Compute the respondent’s annual individual income: $y_0 = y_{\text{HH}} \times 12/k$.
2. Find the respondent’s gpercentile p^* in the IT25 distribution by linear interpolation on the 127-bracket cumulative distribution.
3. Retrieve their 2035 income $\hat{y}_{2035}$ by interpolating the appropriate scenario column of `ineq_IT_2035.csv` at p^* .

The scenario column is determined by the user’s working hours, redistribution scope, decarbonization and public-services choices (after applying δ and ψ corrections to the exported base column).

For hours levels not equal to 35h, a non-convergence scope column is built from the 35h base of *that same scope*, so it keeps the scope’s own within-country shape (national for N, inequality for G) rather than the converged SC shape. The 35h scope base is rescaled to the target hours class x by

$$N_x = SN \frac{\bar{y}_x^I}{SN}, \quad G_x = (SI + \text{cash_GR35}) \frac{\bar{y}_x^C}{SI}, \quad (40)$$

where $\bar{y}_x^I$ ($\bar{y}_x^C$) is the gp-weighted mean of the inequality (convergence) distribution at hours class $x \in \{\text{S15k, B45kC, B90k, P}\}$, and the bars denote 35h-column weighted means in `ineq_IT_2035.csv`. (At $x = 35\text{h}$ the ratio is 1 and the scope base is returned unchanged.)

Point of attention: The IT25 distribution may not be monotone over the full 127 gpercentile range due to imputation noise in the top-1 % brackets. The percentile search is therefore applied only to the 99-bracket bottom distribution; above gpercentile 99, the last bracket’s income is returned.

9 2100 income bar charts

The *national incomes* chart shows the IT 2100 distribution in 21 brackets and the *global incomes* chart shows both the IT and the World 2100 distributions. All values are cash-income-equivalent (scaled by r_{IT}).

For hours variants not equal to 35h, the `ineq_2100` distribution for the given redistribution scope is scaled by:

$$C_{\text{hours}} = \frac{\bar{W}_{\text{hours}}}{\bar{W}_{\text{SC}}}, \quad (41)$$

where $\bar{W}$ denotes the bracket-weighted world average from `ineq_2100.csv`. This ratio equals the hours coef from Table 2 for the GIT+national scope, and generalises to other scopes by the same proportionality.

Point of attention: The bar charts display *average* income within each bracket. The top bracket ($p_{99}p_{100}$) typically shows a very steep increase: in the SC scenario this bracket is compressed by redistribution; in the PI/SI scenarios it can be an order of magnitude above the median. When displaying these charts in the questionnaire, consider truncating or log-scaling the y -axis to avoid the top bracket dominating the visual.

10 Consistency checks and known limitations

1. **SN vs SI:** both sit at the no-global GDP level G^{SI_0} (same average) but differ in shape: SN starts from the nationally-redistributed income (IT35–GR35, i.e. SC’s compressed shape) and SI from the 2025 distribution (IT25). The wedge between them at intermediate percentiles is the effect of national redistribution alone.

2. **SG vs SI:** SG applies a *global* income-tax schedule $\tau^G(\cdot)$ — a function of N income, read off the world SN $\rightarrow$ SC distributions and evaluated at each unit's I (SI) income — to the inequality base ($\text{SG} = \text{SI} (1 - \tau^G(\text{SI}))$). It keeps the no-national-redistribution shape of SI but reallocates income across countries as the global transfer does worldwide: a given income faces the same rate everywhere, negative (subsidy) at low incomes and positive (tax) at high ones. Because τ^G is global, the identity $\text{SG} = \text{SI} (\text{SC}/\text{SN})$ holds at the world level but *not* within IT (IT's percentiles borrow the global income $\rightarrow$ rate map, not IT's own). Hours variants of the G scope scale the base SG by the same hours coef as the other scopes (the income distribution, and hence the rate schedule, scales with the economy).
3. **IT_SC = World_SC at 2100:** Under SC, countries converge to approximately the same per-capita income by 2100, so Italy's 2100 distribution and the world's 2100 distribution should have similar averages. This is confirmed in the data ($\bar{y}_{IT,SC} \approx \bar{y}_{World,SC}$ after applying r_{IT}).
4. **PI base for 2100:** The PI base uses `yp_recomp2025` (pre-GIT national secondary income) rather than `income2025` (which includes the 2025 GIT dividend). Using the post-dividend income would overstate PI incomes for net GIT receivers and understate them for net contributors.
5. **No PI2 scenarios:** The PI scenario does not model sectoral change in [Chancel et al. \(2026\)](#)'s data (`sectoral_change = 0` for all PI variants, regardless of the public-services choice). This is a limitation: PI+expanded PS is computed using structural change = 0, so the temperature does not reflect any emissions benefit from sectoral reallocation.
6. **B90k temperature:** B90k is not in [Chancel et al. \(2026\)](#)'s emission output workbook. Its temperature is predicted by the no-emissions regression, which may not generalise well to the B90k GDP level ($1.5 \times 60 \times 9.41 \approx 847 \text{ T€}$), which lies between SC (565 T€) and PC (1129 T€) but has a distinct productivity path.

11 Data files

ineq_IT_2035.csv 127 rows (`gpercentile`) $\times$ columns (`IT25`, `cash_GR35`, `B90kC`, `SC`, `B90kMat`, `Bmat`, `B45kC`, `B120kC`, `B90kI`, `B90kN`, `B90kG`, `B`, `BG`, `BN`, `BI`, `Bbeef`, `Bflight`, `B90kC_SD`). Income in EUR PPP 2025/adult/yr.

ineq_IT_2035_full.csv 127 rows $\times$ additional columns: all 6 hours classes $\times$ 4 scopes + standalone columns `SCmat`, `B90kMat`, `Bmat`, `Bbeef`, `Bflights` + B90k sub-target variants `B{45k/30k/15k}{C/G/N/I}`. `Bbeef` and `Bflights` equal `B` (income = B

at 2035). B120k{C/G/N/I} uses the 45h-class formula ($\frac{44}{40}\Phi$); B120kG/N/I = B90k{G/N/I} $\times 1.1$.

ineq2_IT_2035.csv, ineq2_IT_2035_full.csv Same as **ineq_IT_2035(_full).csv** but the B scenarios use a 2 % yearly growth for 2025–2035 instead of 1 %. Every B-named column is scaled by $(1.02)^{10}/(1.01)^{10} \approx 1.10$; non-B columns are identical.

ineq_2100.csv 21 rows (brackets) $\times$ columns (bracket label + IT and World for each of the selected scenarios, including B90kC, B120kC). Income in EUR PPP 2025/adult/yr $\times r_{IT}$.

ineq_2100_full.csv 21 rows $\times$ additional columns: all 6 hours classes $\times$ 4 scopes + SCmat + B90k sub-target variants B{45k/30k/15k}{C/G/N/I} + Bbeef, Bflights, Bmat (= SC at 2100, since B=SC at 2100) + B120k{C/G/N/I} (= P{C/G/N/I} at 2100; formerly W{C/G/N/I}).

temp100.csv 54 rows: 30 Chancel et al. (2026) scenarios + 24 new rows for SI/SN/SG/B90k at all decarb $\times$ food combinations. Columns: scenario, gdp (T€), sectoral_change, decarb, temp_final (°C).

conjoint_constants.csv Key-value rows covering: r_{IT} , τ_{PS} , decarbonization factors, temperature adjustment coefficients, GDP per capita by hours $\times$ GIT, population projections, no-emissions regression coefficients, and average incomes from both distribution files (now with B90k/B/B120k column names).

12 Reproducibility

All distributions are generated by `code_simulator/questionnaire.R`, which is self-contained (reads raw Bothe et al. (2026) and Chancel et al. (2026) data files plus `chancel_temp2100_completed.csv`) and produces the four output files above. The conjoint computation at display time is handled by `code_simulator/js/conjoint.js`, which reads the CSV files via `fetch()` and requires a web server.

References

- Bothe, O., Chancel, L., and co-authors (2026). Replication package for Global Justice Report Working Paper 2026-11. *WID.world* Working Paper. URL: <https://wid.world/document/replication-package-working-paper-2026-11/>
- Chancel, L., and co-authors (2026). Global Justice Report 2026. *World Inequality Lab*. Data: <https://wseed.world/data.html>

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